

Program : Diploma in Artificial Intelligence & Machine Learning	
Course Code : 4349	Course Title: Web Application Development using Python Lab
Semester : 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- Students will be able to develop and deploy Web applications using Python frameworks.

Course Prerequisites:

Topic	Course Title	Semester
Basic IT Skills	Introduction to IT systems	1
Knowledge of SQL and its usage	Database lab	3
Python Programming skill	Python Programming Lab	3

Course Outcomes

On completion of the course, students will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Develop interactive web pages	12	Applying
CO2	Develop simple web applications using Python framework	9	Applying
CO3	Develop web applications with multiple forms	12	Applying
CO4	Open ended experiment	9	Applying
	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3	3	3	3	3	3	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Develop interactive web pages		
M1.01	Develop simple web pages using HTML5	3	Applying
M1.02	Apply styles to the web pages using CSS	3	Applying
M1.03	Develop simple web pages using Javascript.	3	Applying
M1.04	Develop simple web pages using jQuery.	3	Applying
CO2	Develop simple web applications using Python framework		
M2.01	Practice installation of Python IDE & Python framework & database server	3	Applying
M2.02	Develop a simple web application using Python framework	3	Applying
M2.03	Develop web applications using Bootstrap in Python framework	3	Applying
	Lab Exam – I	1.5	
CO3	Develop web applications with multiple forms		

M3.01	Develop simple web applications with form.	3	Applying
M3.02	Develop web applications with multiple forms.	3	Applying
M3.03	Demonstrate admin customization	3	Applying
M3.04	Develop web applications for user authentication.	3	Applying
CO4	Open ended experiment		
M4.01	Practice deployment of web applications in a web server or cloud server.	3	Applying
M4.02	Open-ended experiment. **	6	Applying
	Lab Exam – II	1.5	

**** Suggested Open-ended Experiments:**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open-ended experiments as a group of 2-3. There is no duplication in experiments between groups.)

1. Library Management
2. Student Information System
3. Hospital Management

Text / Reference:

T/R	Book Title / Author
T1	Laura Lemay, Rafe Colburn, Jennifer Kyrnin, “ <i>Mastering HTML, CSS & Javascript Web Publishing</i> ”, BPB Publications 2016
T2	Miguel Grinberg, “ <i>Flask Web Development: Developing Web Applications with Python</i> ”, O'Reilly 2018

Online Resources:

Sl. No.	Website Link
1	https://www.tutorialrepublic.com/html-tutorial/
2	https://www.tutorialrepublic.com/css-tutorial/

3	https://www.tutorialrepublic.com/javascript-tutorial/
4	https://www.tutorialrepublic.com/jquery-tutorial/
5	https://www.tutorialrepublic.com/twitter-bootstrap-tutorial/
6	https://www.tutorialrepublic.com/sql-tutorial/
7	https://www.tutorialspoint.com/flask/index.htm
8	https://www.javatpoint.com/flask-tutorial